Lab 3.1 - Mapping the Internet

Objectives

Part 1: Test Network Connectivity Using Ping

Part 2: Trace a Route to a Remote Server Using Windows Tracert

Background

Route tracing computer software is a utility that lists the networks data has to traverse from the user's originating end device to a distant destination network.

This network tool is typically executed at the command line as:

```
tracert <destination network name or end device address>
```

(Microsoft Windows systems)

or

```
traceroute <destination network name or end device address>
```

(UNIX and similar systems)

Route tracing utilities allow a user to determine the path or routes as well as the delay across an IP network. Several tools exist to perform this function.

The **traceroute** (or **tracert**) tool is often used for network troubleshooting. By showing a list of routers traversed, it allows the user to identify the path taken to reach a particular destination on the network or across internetworks. Each router represents a point where one network connects to another network and through which the data packet was forwarded. The number of routers is known as the number of "hops" the data traveled from source to destination.

The displayed list can help identify data flow problems when trying to access a service such as a website. It can also be useful when performing tasks such as downloading data. If there are multiple websites (mirrors) available for the same data file, one can trace each mirror to get a good idea of which mirror would be the fastest to use.

Two trace routes between the same source and destination conducted some time apart may produce different results. This is due to the "meshed" nature of the interconnected networks that comprise the Internet and the Internet Protocols ability to select different pathways over which to send packets.

Command-line-based route tracing tools are usually embedded with the operating system of the end device.

Scenario

Using an Internet connection, you will use three route tracing utilities to examine the Internet pathway to destination networks. This activity should be performed on a computer that has Internet access and access to the command line. First, you will use the Windows embedded tracert utility.

Required Resources

1 PC (Windows 7 or 8 with Internet access)

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Part 1: Test Network Connectivity Using Ping

Step 1: Determine whether the remote server is reachable.

To trace the route to a distant network, the PC used must have a working connection to the Internet.

a. The first tool we will use is ping. Ping is a tool used to test whether a host is reachable. Packets of information are sent to the remote host with instructions to reply. Your local PC measures whether a response is received to each packet, and how long it takes for those packets to cross the network. The name ping comes from active sonar technology in which a pulse of sound is sent underwater and bounced off of terrain or other ships.

b. From your PC, click the **Windows Start** icon, type **cmd** in the **Search programs and files** box, and then press Enter.

c. At the command-line prompt, type **ping www.cisco.com**.

```
C:\>ping www.cisco.com

Pinging e144.dscb.akamaiedge.net [72.246.234.230] with 32 bytes of data:
Reply from 72.246.234.230: bytes=32 time=3ms TTL=53
Reply from 72.246.234.230: bytes=32 time=2ms TTL=53
Reply from 72.246.234.230: bytes=32 time=2ms TTL=53
Reply from 72.246.234.230: bytes=32 time=2ms TTL=53

Ping statistics for 72.246.234.230:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

d. The first output line displays the Fully Qualified Domain Name (FQDN) e144.dscb.akamaiedge.net. This is followed by the IP address 72.246.234.230. Cisco hosts the same web content on different servers throughout the world (known as mirrors). Therefore, depending upon where you are geographically, the FQDN and the IP address will be different.

e. From this portion of the output:

```
Ping statistics for 72.246.234.230:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

Four pings were sent and a reply was received from each ping. Because each ping was responded to, there was 0% packet loss. On average, it took 2 ms (2 milliseconds) for the packets to cross the network. A millisecond is 1/1,000th of a second.

Streaming video and online games are two applications that suffer when there is packet loss, or a slow network connection. A more accurate determination of an Internet connection speed can be determined by sending 100 pings, instead of the default 4. Here is how to do that:

```
ping -n 100 cisco.com
```

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And here is what the output from that looks like:

```
Ping statistics for 72.246.234.230:
    Packets: Sent = 100, Received = 100, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 2ms
```

f. Now ping Regional Internet Registry (RIR) websites located in different parts of the world:

For Africa:

`ping www.afrinic.net`

```
↳ ping www.afrinic.net
PING www.afrinic.net (196.216.3.4) 56(84) bytes of data.
64 bytes from lb.iso.afrinic.net (196.216.3.4): icmp_seq=1 ttl=46 time=300 ms
64 bytes from lb.iso.afrinic.net (196.216.3.4): icmp_seq=2 ttl=46 time=300 ms
64 bytes from lb.iso.afrinic.net (196.216.3.4): icmp_seq=3 ttl=46 time=300 ms
64 bytes from lb.iso.afrinic.net (196.216.3.4): icmp_seq=4 ttl=46 time=300 ms
64 bytes from lb.iso.afrinic.net (196.216.3.4): icmp_seq=5 ttl=46 time=300 ms
^C
--- www.afrinic.net ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 299.871/300.222/300.458/0.203 ms
```

For Asia-Pacific:

`ping www.apnic.net`

```
↳ ping www.apnic.net
PING www.apnic.net.cdn.cloudflare.net (104.18.235.68) 56(84) bytes of data.
64 bytes from 104.18.235.68: icmp_seq=1 ttl=46 time=27.7 ms
64 bytes from 104.18.235.68: icmp_seq=2 ttl=46 time=28.0 ms
64 bytes from 104.18.235.68: icmp_seq=3 ttl=46 time=27.5 ms
64 bytes from 104.18.235.68: icmp_seq=4 ttl=46 time=28.3 ms
64 bytes from 104.18.235.68: icmp_seq=5 ttl=46 time=27.8 ms
^C
--- www.apnic.net.cdn.cloudflare.net ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 27.520/27.854/28.252/0.245 ms
```

For Europe:

`ping www.ripe.net`

```
↳ ping www.ripe.net
PING www.ripe.net.cdn.cloudflare.net (104.18.5.245) 56(84) bytes of data.
64 bytes from 104.18.5.245: icmp_seq=1 ttl=46 time=29.2 ms
64 bytes from 104.18.5.245: icmp_seq=2 ttl=46 time=29.4 ms
64 bytes from 104.18.5.245: icmp_seq=3 ttl=46 time=30.5 ms
64 bytes from 104.18.5.245: icmp_seq=4 ttl=46 time=30.4 ms
64 bytes from 104.18.5.245: icmp_seq=5 ttl=46 time=29.7 ms
^C
--- www.ripe.net.cdn.cloudflare.net ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4003ms
rtt min/avg/max/mdev = 29.182/29.831/30.479/0.517 ms
```

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For South America:

```
ping www.lacnic.net
```

```
↳ ping www.lacnic.net
PING www.lacnic.net (200.3.14.145) 56(84) bytes of data.
64 bytes from www.lacnic.net (200.3.14.145): icmp_seq=1 ttl=43 time=376 ms
64 bytes from www.lacnic.net (200.3.14.145): icmp_seq=2 ttl=43 time=377 ms
64 bytes from www.lacnic.net (200.3.14.145): icmp_seq=3 ttl=43 time=376 ms
64 bytes from www.lacnic.net (200.3.14.145): icmp_seq=4 ttl=43 time=376 ms
64 bytes from www.lacnic.net (200.3.14.145): icmp_seq=5 ttl=43 time=377 ms
^C
--- www.lacnic.net ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4001ms
rtt min/avg/max/mdev = 376.061/376.346/376.628/0.229 ms
```

All these pings were run from a computer located in Thailand. What happens to the average ping time in milliseconds when data is traveling within the same continent (Asia) as compared to data from Asia traveling to different continents?

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What is interesting and unique about the pings that were sent to Afirca and South America's website (based on given figure above)?

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Part 2: Trace a Route to a Remote Server Using Tracert

Step 1: Determine what route across the Internet traffic takes to the remote server.

Now that basic reachability has been verified by using the ping tool, it is helpful to look more closely at each network segment that is crossed. To do this, the **tracert** tool will be used.

a. At the command-line prompt, type **tracert www.cisco.com**.

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```
C:\>tracert www.cisco.com

Tracing route to e144.dscb.akamaiedge.net [72.246.234.230]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.10.254
  2    <1 ms    <1 ms    <1 ms  161.200.92.62
  3     *       <1 ms    <1 ms  161.200.93.254
  4     8 ms     2 ms     1 ms  161.200.83.129
  5    <1 ms    <1 ms    <1 ms  f1-0-2-8510-cen32.it.chula.ac.th [161.20
4]
  6    <1 ms    <1 ms    <1 ms  161.200.255.94
  7     1 ms     1 ms     1 ms  161.200.255.214
  8    88 ms     3 ms     2 ms  mx-ll-110-164-51-177.static.3bb.co.th [1
1.177]
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     4 ms     3 ms     3 ms  mx-ll-110.164.0-59.static.3bb.co.th [110
9]
 12     3 ms     3 ms     2 ms  mx-ll-110.164.14-192.static.3bb.co.th [1
4.192]
 13     2 ms     2 ms     2 ms  a72-246-234-230.deploy.akamaitechnologie
2.246.234.230]

Trace complete.
```

b. Save the tracert output in a text file as follows:

   1) Right-click the title bar of the Command Prompt window and choose **Edit** > **Select All**.

   2) Right-click the title bar of the Command Prompt window again and choose **Edit** > **Copy**.

   3) Open the **Windows Notepad** program: **Windows Start** icon > **All Programs** > **Accessories** > **Notepad**.

   4) To paste the output into Notepad, choose **Edit** > **Paste**.

   5) Choose **File** > **Save As** and save the Notepad file to your desktop as **tracert1.txt**.

c. Run **tracert** for each destination website and save the output in sequentially numbered files.

```
> tracert www.afrinic.net
  tracert www.lacnic.net
```

d. Interpreting **tracert** outputs.

   Routes traced can go through many hops and a number of different Internet Service Providers (ISPs), depending on the size of your ISP, and the location of the source and destination hosts. Each “hop” represents a router. A router is a specialized type of computer used to direct traffic across the Internet. Imagine taking an automobile trip across several countries using many highways. At different points in the trip, you come to a fork in the road in which you have the option to select from several different highways. Now further imagine that there is a device at each fork in the road that directs you to take the correct highway to your final destination. That is what a router does for packets on a network.

   Because computers talk in numbers, rather than words, routers are uniquely identified using IP addresses (numbers with the format x.x.x.x). The **tracert** tool shows you what path through the network a packet of information takes to reach its final destination. The **tracert** tool also gives you an idea of how fast traffic is going on each segment of the network. Three packets are sent to each router in the path, and the return time is measured in milliseconds. Now use this information to analyze the **tracert** results to www.cisco.com. Below is the entire traceroute:

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```
C:\>tracert www.cisco.com

Tracing route to e144.dscb.akamaiedge.net [23.1.144.170]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  dslrouter.westell.com [192.168.1.1]
  2    38 ms    38 ms    37 ms  10.18.20.1
  3    37 ms    37 ms    37 ms  G3-0-9-2204.ALBYNY-LCR-02.verizon-gni.net [130.8
1.196.190]
  4    43 ms    43 ms    42 ms  so-5-1-1-0.NY325-BB-RTR2.verizon-gni.net [130.81
.22.46]
  5    43 ms    43 ms    65 ms  0.so-4-0-2.XT2.NYC4.ALTER.NET [152.63.1.57]
  6    45 ms    45 ms    45 ms  0.so-3-2-0.XL4.EWR6.ALTER.NET [152.63.17.109]
  7    46 ms    48 ms    46 ms  TenGigE0-5-0-0.GW8.EWR6.ALTER.NET [152.63.21.14]

  8    45 ms    45 ms    45 ms  a23-1-144-170.deploy.akamaitechnologies.com [23.
1.144.170]

Trace complete.
```

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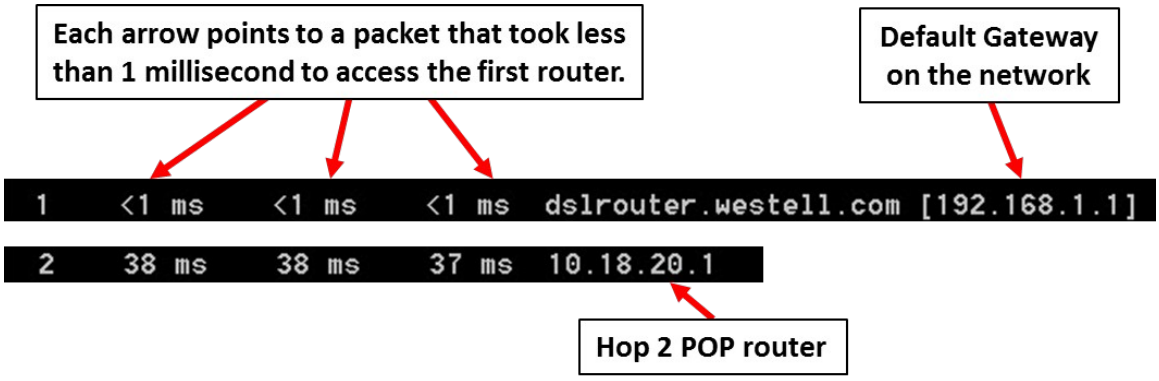
Below is the breakdown:



In the example output shown above, the tracert packets travel from the source PC to the local router default gateway (hop 1: 192.168.1.1) to the ISPs Point of Presence (POP) router (hop 2: 10.18.20.1). Every ISP has numerous POP routers. These POP routers are at the edge of the ISP's network and are the means by which customers connect to the Internet. The packets travel along the Verizon network for two hops and then jump to a router that belongs to alter.net. This could mean that the packets have traveled to another ISP. This is significant because sometimes there is packet loss in the transition between ISPs, or sometimes one ISP is slower than another. How could we determine if alter.net is another ISP or the same ISP?

e. There is an Internet tool known as whois. The whois tool allows us to determine who owns a domain name. A web-based whois tool is found at http://whois.domaintools.com/. This domain is also owned by Verizon according to the web-based whois tool.

```
Registrant:
        Verizon Business Global LLC
        Verizon Business Global LLC
        One Verizon Way
         Basking Ridge NJ 07920
        US
         domainlegalcontact@verizon.com +1.7033513164 Fax: +1.7033513669

    Domain Name: alter.net
```

To summarize, Internet traffic starts at a home PC and travels through the home router (hop 1). It then connects to the ISP and travels through its network (hops 2-7) until it arrives at the remote server (hop 8). This is a relatively unusual example in which there is only one ISP involved from start to finish. It is typical to have two or more ISP involved as displayed in the following examples.

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f. Now examine an example that involves Internet traffic crossing multiple ISPs. Below is the tracert for www.afrinic.net：

```
C:\>tracert www.afrinic.net

Tracing route to www.afrinic.net [196.216.2.136]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms  dslrouter.westell.com [192.168.1.1]
  2    39 ms    38 ms    37 ms  10.18.20.1
  3    40 ms    38 ms    39 ms  G4-0-0-2204.ALBYNY-LCR-02.verizon-gni.net [130.8
1.197.182]
  4    44 ms    43 ms    43 ms  so-5-1-1-0.NY325-BB-RTR2.verizon-gni.net [130.81
.22.46]
  5    43 ms    43 ms    42 ms  0.so-4-0-0.XT2.NYC4.ALTER.NET [152.63.9.249]
  6    43 ms    71 ms    43 ms  0.ae4.BR3.NYC4.ALTER.NET [152.63.16.185]
  7    47 ms    47 ms    47 ms  te-7-3-0.edge2.NewYork2.level3.net [4.68.111.137
]
  8    43 ms    55 ms    43 ms  vlan51.ebr1.NewYork2.Level3.net [4.69.138.222]
  9    52 ms    51 ms    51 ms  ae-3-3.ebr2.Washington1.Level3.net [4.69.132.89]

 10   130 ms   132 ms   132 ms  ae-42-42.ebr2.Paris1.Level3.net [4.69.137.53]
 11   139 ms   145 ms   140 ms  ae-46-46.ebr1.Frankfurt1.Level3.net [4.69.143.13
7]
 12   148 ms   140 ms   152 ms  ae-91-91.csw4.Frankfurt1.Level3.net [4.69.140.14
]
 13   144 ms   144 ms   146 ms  ae-92-92.ebr2.Frankfurt1.Level3.net [4.69.140.29
]
 14   151 ms   150 ms   150 ms  ae-23-23.ebr2.London1.Level3.net [4.69.148.193]

 15   150 ms   150 ms   150 ms  ae-58-223.csw2.London1.Level3.net [4.69.153.138]

 16   156 ms   156 ms   156 ms  ae-227-3603.edge3.London1.Level3.net [4.69.166.1
54]
 17   157 ms   159 ms   160 ms  195.50.124.34
 18   353 ms   340 ms   341 ms  168.209.201.74
 19   333 ms   333 ms   332 ms  csw4-pkl-gi1-1.ip.isnet.net [196.26.0.101]
 20   331 ms   331 ms   331 ms  196.37.155.180
 21   318 ms   316 ms   318 ms  fa1-0-1.ar02.jnb.afrinic.net [196.216.3.132]
 22   332 ms   334 ms   332 ms  196.216.2.136

Trace complete.
```

What happens at hop 7? Is level3.net the same ISP as hops 2-6, or a different ISP? Use the whois tool to answer this question.

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What happens in hop 10 to the amount of time it takes for a packet to travel between Washington D.C. and Paris, as compared with the earlier hops 1-9?

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What happens in hop 18? Do a whois lookup on 168.209.201.74 using the whois tool. Who owns this network?

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g. Type **tracert www.lacnic.net**.

```
C:\>tracert www.lacnic.net

Tracing route to www.lacnic.net [200.3.14.147]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  dslrouter.westell.com [192.168.1.1]
  2    38 ms    38 ms    37 ms  10.18.20.1
  3    38 ms    38 ms    39 ms  G3-0-9-2204.ALBYNY-LCR-02.verizon-gni.net [130.8
1.196.190]
  4    42 ms    43 ms    42 ms  so-5-1-1-0.NY325-BB-RTR2.verizon-gni.net [130.81
.22.46]
  5    82 ms    47 ms    47 ms  0.ae2.BR3.NYC4.ALTER.NET [152.63.16.49]
  6    46 ms    47 ms    56 ms  204.255.168.194
  7   157 ms   158 ms   157 ms  ge-1-1-0.100.gw1.gc.registro.br [159.63.48.38]
  8   156 ms   157 ms   157 ms  xe-5-0-1-0.core1.gc.registro.br [200.160.0.174]

  9   161 ms   161 ms   161 ms  xe-4-0-0-0.core2.nu.registro.br [200.160.0.164]

 10   158 ms   157 ms   157 ms  ae0-0.ar3.nu.registro.br [200.160.0.249]
 11   176 ms   176 ms   170 ms  gw02.lacnic.registro.br [200.160.0.213]
 12   158 ms   158 ms   158 ms  200.3.12.36
 13   157 ms   158 ms   157 ms  200.3.14.147

Trace complete.
```

What happens in hop 7?

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Reflection

What are the functional differences between the commands ping and tracert?

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